Welcome to Mars.
Please Don’t Die

Laboratory of Advanced Programming - Hackathon exam

Matteo Mastranza - 1971730
Camille Eugenio - 2216495
Gianluca Sanfilippo - 1943663

SAPIENZA
UNIVERSITÀ DI ROMA

Master’s in Engineering in Computer Science - A.Y. 2025/2026

Table of contents

1 Our mission

2 Architecture

3 Technology stack

4 Containers

5 User stories

6 Demo

Our Mission

Ingest

Collect heterogeneous data from 15 sensors (8 REST polling and 7 telemetry streams)

Normalize

Convert raw data into a standardized internal representation

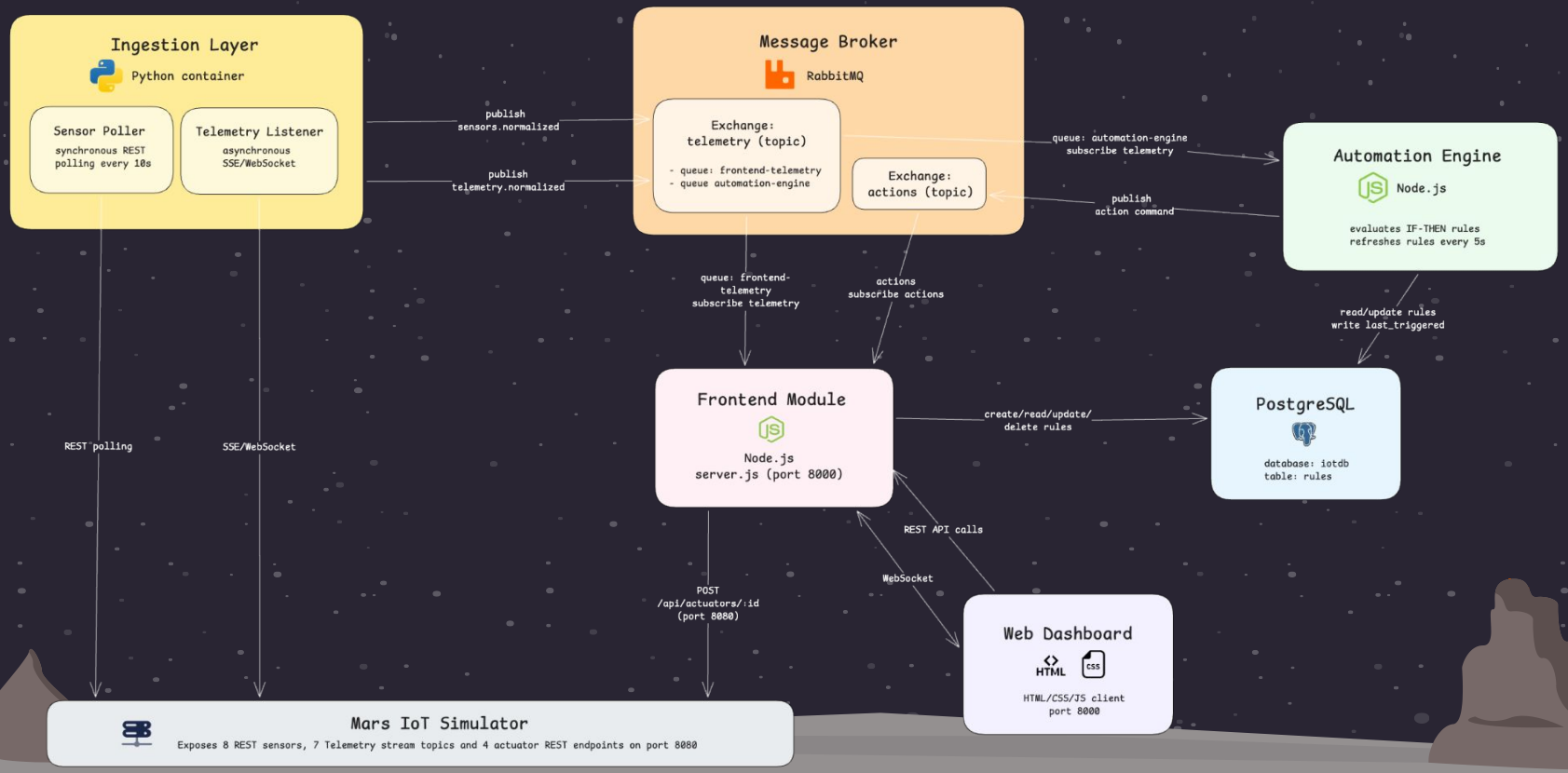
Automate

Evaluate simple automation rules to manage environmental control via actuators

Monitor

Provide a real-time dashboard for monitoring sensors and telemetry data and toggle actuators

System Architecture



Technology Stack

Python

Ingestion & Normalization

Node.js

Automation Engine & Frontend

RabbitMQ

Message Broker

PostgreSQL

Rule persistence

Docker

Containerization

HTML, CSS & JS

Web UI

Containers Overview

sensor_poller
REST polling every 10s
→ sensors.normalized

telemetry_listener
SSE streams
→ telemetry.normalized

rabbitmq
Event bus
Topic exchanges

automation-engine
IF-THEN rules evaluation

frontend_module
REST API + WebSocket gateway

postgres
Rule persistence

Data ingestion and Standardization

Rest polling

SSE stream

publish sensors.normalized

publish telemetry..normalized

Sensors

Two different sources:
8 REST sensors + 7 telemetry streams

Normalization

From heterogeneous formats to a single unified event schema

{"timestamp": "2026-03-08T09:31:44.803625+00:00", "sensor_id": "greenhouse_temperature", "metric": "temperature_c", "value": 24.09, "unit": "C", "status": "ok"}

Message Broker

Routes normalized events to subscribers
Exchange: telemetry

Automation engine

Subscribe

Consumes telemetry exchange from RabbitMQ

Evaluate

Compares incoming sensor data against enabled IF-THEN rules

Rules refreshed from PostgreSQL every 5s

Publish

Sends action command to RabbitMQ actions exchange

Example

IF co2_hall.co2_ppm > 700
THEN set hall_ventilation to ON

→ publishes to actions exchange
→ hall_ventilation set ON

Frontend Module

Subscribes to RabbitMQ (telemetry + actions)

Maintains in memory **sensor cache**

Forwards actuator commands to simulator via REST

Serves static dashboard on port 8000

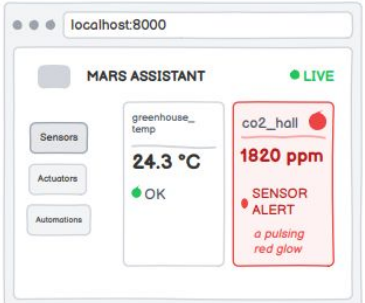
Real-time Dashboard

- Single Page Application: built with JS, HTML and CSS with 3 sections (Sensors, Actuators, Automations)
- Real-Time: updated with WebSockets
- Allows users to **monitor sensors**, **toggle actuators** and **manage rules**

Some User Stories

US-S3

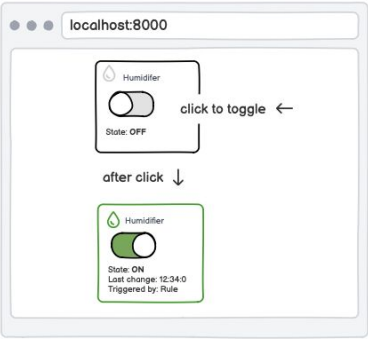
Be notified when sensor status is "warning"



↑ Widget pulses red with "!" badge and SENSOR ALERT status when status=warning/critical

US-A6

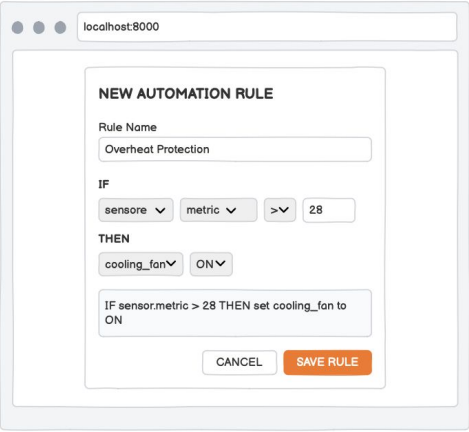
Toggle actuator state directly from widget



↑ Click toggle -> immediate state change, no separate page needed

US-AU9

Create automation rule with IF-THEN interface



↑ Modal with IF-THEN form, live preview of rule, sensor+metric dropdowns

DEMO

Thank you for you attention